

19_Gaussian Elimination

1. What is Gaussian Elimination?

Gaussian Elimination is a method for solving a system of linear equations by converting the system's augmented matrix into **row-echelon form (upper triangular form)** using **elementary row operations**, and then solving by **back-substitution**.

Elementary Row Operations:

1. Swap two rows.
2. Multiply a row by a nonzero scalar.
3. Add or subtract a multiple of one row to another row.

2. Steps of Gaussian Elimination

1. **Write the augmented matrix** of the system of equations.
2. **Use row operations** to get zeros below the leading 1 in the first column (pivot column).
3. Move to the next column and **repeat** the process for the submatrix.
4. Continue until the matrix is in **row-echelon form**.
5. **Back-substitute** to find the solution.

Given System of Linear Equations

$$\begin{array}{lcl} (1) & 2x + y - z & = 8 \\ (2) & -3x - y + 2z & = -11 \\ (3) & -2x + y + 2z & = -3 \end{array}$$

Given system (augmented matrix)

$$\left[\begin{array}{ccc|c} 2 & 1 & -1 & 8 \\ -3 & -1 & 2 & -11 \\ -2 & 1 & 2 & -3 \end{array} \right]$$

Step 1: Make the first pivot equal to 1

Divide **Row 1** by 2:

$$R_1 \rightarrow \frac{1}{2}R_1$$
$$\left[\begin{array}{ccc|c} 1 & \frac{1}{2} & -\frac{1}{2} & 4 \\ -3 & -1 & 2 & -11 \\ -2 & 1 & 2 & -3 \end{array} \right]$$

Step 2: Eliminate x from Row 2

Add $3R_1$ to Row 2:

$$R_2 \rightarrow R_2 + 3R_1$$

Calculation:

- $-3 + 3(1) = 0$
- $-1 + 3\left(\frac{1}{2}\right) = \frac{1}{2}$
- $2 + 3\left(-\frac{1}{2}\right) = \frac{1}{2}$
- $-11 + 3(4) = 1$

$$\left[\begin{array}{ccc|c} 1 & \frac{1}{2} & -\frac{1}{2} & 4 \\ 0 & \frac{1}{2} & \frac{1}{2} & 1 \\ -2 & 1 & 2 & -3 \end{array} \right]$$

Step 3: Eliminate x from Row 3

Add $2R_1$ to Row 3:

$$R_3 \rightarrow R_3 + 2R_1$$

Calculation:

- $-2 + 2(1) = 0$
- $1 + 2\left(\frac{1}{2}\right) = 2$
- $2 + 2\left(-\frac{1}{2}\right) = 1$
- $-3 + 2(4) = 5$

$$\left[\begin{array}{ccc|c} 1 & \frac{1}{2} & -\frac{1}{2} & 4 \\ 0 & \frac{1}{2} & \frac{1}{2} & 1 \\ 0 & 2 & 1 & 5 \end{array} \right]$$

Step 4: Make the second pivot equal to 1

Multiply Row 2 by 2:

$$R_2 \rightarrow 2R_2$$

$$\left[\begin{array}{ccc|c} 1 & \frac{1}{2} & -\frac{1}{2} & 4 \\ 0 & 1 & 1 & 2 \\ 0 & 2 & 1 & 5 \end{array} \right]$$

Step 5: Eliminate y from Row 3

Subtract $2R_2$ from Row 3:

$$R_3 \rightarrow R_3 - 2R_2$$

Calculation:

- $2 - 2 = 0$
- $1 - 2 = -1$
- $5 - 4 = 1$

$$\left[\begin{array}{ccc|c} 1 & \frac{1}{2} & -\frac{1}{2} & 4 \\ 0 & 1 & 1 & 2 \\ 0 & 0 & -1 & 1 \end{array} \right]$$

Step 6: Back substitution

From Row 3:

$$-z = 1 \Rightarrow z = -1$$

From Row 2:

$$y + z = 2 \Rightarrow y = 3$$

From Row 1:

$$x + \frac{1}{2}y - \frac{1}{2}z = 4$$

$$x + \frac{3}{2} + \frac{1}{2} = 4 \Rightarrow x = 2$$

 **Final Answer**

$$\boxed{x = 2, y = 3, z = -1}$$

Example 02:

Solve the system:

$$\begin{cases} 2x + y - z = 1 \\ -3x - y + 2z = -4 \\ x + 2y - z = 0 \end{cases}$$

Step 1: Write the augmented matrix

$$\left[\begin{array}{ccc|c} 2 & 1 & -1 & 1 \\ -3 & -1 & 2 & -4 \\ 1 & 2 & -1 & 0 \end{array} \right]$$

Step 2: Make leading 1 in first row

We can divide the first row by 2 to make it easier, but sometimes it's optional. Here, I'll leave it as is and eliminate below.

Eliminate first column below pivot (2 in R₁):

- $R_2 \rightarrow R_2 + \frac{3}{2}R_1$
- $R_3 \rightarrow R_3 - \frac{1}{2}R_1$

Compute $R_2 + \frac{3}{2}R_1$

$$\begin{aligned} R_1 &= [2, 1, -1, 1], \\ \frac{3}{2}R_1 &= [3, 1.5, -1.5, 1.5] \end{aligned}$$

Add to $R_2 = [-3, -1, 2, -4]$:

- First element: $-3 + 3 = 0$
- Second element: $-1 + 1.5 = 0.5$
- Third element: $2 + (-1.5) = 0.5$
- Fourth element: $-4 + 1.5 = -2.5$

So:

$$R_2 = [0, 0.5, 0.5, -2.5] \text{(correct!)}$$

Compute $R_3 - \frac{1}{2}R_1$

$$\frac{1}{2}R_1 = [1, 0.5, -0.5, 0.5]$$

Subtract from $R_3 = [1, 2, -1, 0]$:

- First element: $1 - 1 = 0$
- Second element: $2 - 0.5 = 1.5$
- Third element: $-1 - (-0.5) = -1 + 0.5 = -0.5$
- Fourth element: $0 - 0.5 = -0.5$

So:

$$R_3 = [0, 1.5, -0.5, -0.5] \text{(also correct!)}$$

New matrix:

$$\left[\begin{array}{ccc|c} 2 & 1 & -1 & 1 \\ 0 & 0.5 & 0.5 & -2.5 \\ 0 & 1.5 & -0.5 & -0.5 \end{array} \right]$$

Step 3: Make pivot in second row = 1

Divide R_2 by 0.5:

$$R_2 \rightarrow [0, 1, 1, -5]$$

Matrix:

$$\left[\begin{array}{ccc|c} 2 & 1 & -1 & 1 \\ 0 & 1 & 1 & -5 \\ 0 & 1.5 & -0.5 & -0.5 \end{array} \right]$$

Step 4: Eliminate above and below pivot in column 2

- $R_1 \rightarrow R_1 - 1 \cdot R_2$
- $R_3 \rightarrow R_3 - 1.5 \cdot R_2$

$$R_1: [2 - 0, 1 - 1, -1 - 1, 1 - (-5)] = [2, 0, -2, 6]$$

$$R_3: [0 - 0, 1.5 - 1.5, -0.5 - 1.5, -0.5 - (-7.5)] = [0, 0, -2, 7]$$

New matrix:

$$\left[\begin{array}{ccc|c} 2 & 0 & -2 & 6 \\ 0 & 1 & 1 & -5 \\ 0 & 0 & -2 & 7 \end{array} \right]$$

Step 5: Make pivot in third row = 1

$$R_3 \rightarrow R_3 / -2 = [0, 0, 1, -3.5]$$

Matrix:

$$\left[\begin{array}{ccc|c} 2 & 0 & -2 & 6 \\ 0 & 1 & 1 & -5 \\ 0 & 0 & 1 & -3.5 \end{array} \right]$$

Step 6: Back-substitution

- From R_3 : $z = -3.5$
- From R_2 : $y + z = -5 \Rightarrow y - 3.5 = -5 \Rightarrow y = -1.5$
- From R_1 : $2x - 2z = 6 \Rightarrow 2x - 2(-3.5) = 6 \Rightarrow 2x + 7 = 6 \Rightarrow x = -0.5$

✓ Solution

$$x = -0.5, y = -1.5, z = -3.5$$